

EMPLOYEE TABLE:

Emp ID	FNAME	LNAME	Salary	Dept ID
1	Ananya	Gupta	20000	5
2	Rohan	Kapoor	25000	3
3	Meera	Nair	45000	8
4	Arjun	Singh	85000	2
5	Kavya	Iyer	50000	3

DEPARTMENT TABLE:

Dept ID	Dept Name	Location
1	HR	Bengaluru
2	Sales	Mysore
3	IT	Hyderabad

Relational Algebra Expressions

1. Employees in department ID 3

$\pi_{FNAME, LNAME}(\sigma_{DeptID=3}(EMPLOYEE))$

FNAME	LNAME
Rohan	Kapoor
Kavya	Iyer

2. Employees with salary > 30000

$\pi_{FNAME, Salary}(\sigma_{Salary>30000}(EMPLOYEE))$

FNAME	Salary
Meera	45000
Arjun	85000
Kavya	50000

3. Department name located in Bengaluru

$\pi_{\text{DeptName}}(\sigma_{\text{Location}=\text{"Bengaluru"}}(\text{DEPARTMENT}))$

Dept Name
HR

4. Employees in dept ID:2 and with salary>40000

$\pi_{\text{FNAME}, \text{Salary}}(\sigma_{\text{DeptID}=2 \text{ AND } \text{Salary}>40000}(\text{EMPLOYEE}))$

FNAME	Salary
Arjun	85000

5. Employees with salary<30000

$\pi_{\text{EmpID}, \text{FNAME}}(\sigma_{\text{Salary}<30000}(\text{EMPLOYEE}))$

Emp ID	FNAME
5	Ananya
3	Rohan